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Ms. S. Sreelakshmi
Data Administrator, Elsevier Production
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Dr. Robert Prinz
Handling Editor, *BioSystems*
Elsevier B.V.

Re: BIO 105919 — Article reference BIO_BIOSYS-D-26-00689
"Robust error-minimization in the genetic code across physicochemical metrics and variant codes: a graph-theoretic analysis in GF(2)^6"

Dear Ms. Sreelakshmi and Dr. Prinz,

Thank you for accepting our article for publication in *BioSystems* and for your patience while we resolve the electronic-file issue flagged by production. We are grateful for the care your team is taking with the typesetting.

Nature of the replacement. We are supplying a corrected submission bundle. The bundle addresses two categories of change simultaneously:

1. The immediate production issue: the previously supplied source PDF lacked figures and the reference list. The replacement `manuscript.tex` and `supplement.tex` embed every figure and render the complete reference list; both compile self-contained under pdflatex + elsarticle.
2. A pre-publication QA/QC pass carried out between acceptance and typesetting. Because that pass revised several displayed values, provenance statements, and framing sentences, the enclosed files contain post-acceptance author corrections in addition to the electronic-file repair. We therefore respectfully request the Handling Editor's approval of the revised accepted files. The full accepted-to-final diff is provided as `corrections_ledger.pdf` (attached); every user-visible change is enumerated there with its old value, new value, and justification.

What did not change. The scientific conclusions and the claim hierarchy are unchanged. The four SUPPORTED headline findings (cross-metric coloring optimality, per-table preservation across the 27 NCBI translation tables, ρ-robustness, and $\mathbf{H}(3, 4)$ topology-avoidance depletion), the FALSIFIED KRAS–Fano prediction, the EXPLORATORY tRNA enrichment classification, and the three REJECTED / two TAUTOLOGICAL entries all retain their accepted status. The hierarchy total remains 15 claims.

What was corrected. The most consequential corrections are: (a) a Monte Carlo tie-inclusion fix in the coloring-optimality null (Grantham p moves 0.006099 to 0.006199; other three metrics unchanged); (b) a reconciliation of the *S. cerevisiae* and *Y. lipolytica* mitochondrial tRNA vectors against primary literature (Bonitz *et al.* 1980 and Kerscher *et al.* 2001, both added to the bibliography); (c) a full Bron–Kerbosch MIS enumeration for the tRNA-enrichment robustness bound (the enumeration now returns 332 MIS of size 6 rather than 2, and the per-MIS Stouffer distribution replaces the two-point best/worst summary); (d) a rewrite of the main-text § 2.3.6 tRNA analysis population paragraph to match the 24-row provenance table row-for-row; (e) a narrowing of the primary *H(3, 4)* clade-exclusion statement in § 3.4 to $p < 10^{-3}$ (largest $p \approx 2 \times 10^{-4}$), with the Q_6/new-disconnection sensitivity retaining $p < 10^{-5}$; and (f) a rewrite of Supplement § S18 as a descriptive conservative-threshold sensitivity rather than a formal cross-family Bonferroni correction (which is not well-defined across mixed hypothesistest / model-selection statistics). None of these corrections change any claim's status, and the ledger states each before/after value exactly.

Reproducibility. Both the corrected PDFs and every displayed statistic are reproducible end-to-end from the public repository at tag `v0.6.1` by one canonical command:

```
git clone https://github.com/biostochastics/codontopo && cd codontopo
git checkout v0.6.1
pip install -e ".[dev]"
bash scripts/build_publisher_release.sh
```

The wrapper script `build_publisher_release.sh` runs the analysis pipeline (`codon-topo all` at seed 135325, **n** = 10 000), the provenance-emit and table-generation scripts, both R figure scripts, and the two Typst PDF compilations, in that order. The same command block is quoted verbatim in `CHANGELOG.md`, in `README.md`, in the manuscript Data-and-code availability paragraph, and in Supplement §S24, so no ambiguity remains about which recipe reproduces the paper. Python dependency versions are pinned in `requirements.lock` and a reference build container is provided by `Dockerfile` at the repository root. A SHA-256 manifest of the enclosed release artefacts is included as `MANIFEST.sha256`.

Further review. If your copy-editors or the editorial office would like us to prepare any additional derivatives — for instance a single-column proof-ready PDF, a separate figure-only PDF, individual TIFFs, or a merged manuscript-plus-supplement file — please let us know and we will supply them promptly. We are equally happy to answer any question that arises during typesetting or copy-editing, either by email or by a brief call.

Thank you again for the opportunity to publish this work in *BioSystems* and for your patience while we prepared the corrected files.

With kind regards, on behalf of both authors,

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Enclosures.
1. `manuscript.pdf` — corrected self-contained PDF, all figures and references embedded.
2. `supplement.pdf` — supplementary material, self-contained.
3. `submission_bundle.zip` — corrected LaTeX source archive: `manuscript.tex`, `supplement.tex`, `elsarticle.cls`, `references.bib`, figure PNGs and TIFFs at 300 DPI, and the pdflatex-compiled PDFs.
4. `submission_em_v0.6.1/` — unpacked source of the same LaTeX bundle for reviewers preferring an unarchived layout.
5. `corrections_ledger.pdf` — pre-publication QA/QC accepted-to-final diff ledger.
6. `elsevier_response.pdf` — this letter.
7. `CHANGELOG.md` — consolidated release notes for the v0.6.1 release.
8. `2026-08-14-corrections-ledger.md` — Markdown source of the ledger.
9. `MANIFEST.sha256` — SHA-256 hashes of every enclosed file for integrity verification.